Exploring Structural Phase Transitions in Language with Assembly Theory

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Understanding how and when language reflects the mechanism for its selection remains a central challenge for language-evolution research. Despite a rich lineage of information-theoretic and complex-systems approaches to language (Altmann & Gerlach, 2016), many metrics still struggle to separate evolved, adaptive structures from stochastic variability, and often rely on computability or description-length formalisms rather than how objects are physically constructed and empirically manifest in observed systems. Assembly Theory (AT) offers a new framework to interrogate evolutionary patterns by quantifying the complexity of an object using two empirical measures: the assembly index (a)—the minimum number of recursive steps required to combinatorically construct an object from elementary units—and the copy number (n)—the number of observed identical copies of the object within a set of N objects (Sharma et al., 2023). This approach has been shown to capture signatures of selection and evolution across diverse systems; from organic chemistry to minerals, with new insights for origins-of-life research and biosignature science (Marshall et al., 2021; Patarroyo et al., 2025). It is built on the premise that complex objects (high-a) are exponentially unlikely to appear multiple times by chance, as captured by a quantity called Assembly, defined as $A = \sum_i^N e^{a_i} \frac{n_i - 1}{N}$. Applied to language, systems characterized by high A are expected to provide evidence of enhanced recursivity, hierarchical modularity, and selective forces to structure information into a repeated, complex object. This constructability-and-abundance perspective differs from compression metrics (e.g., Ziv & Lempel, 1978) or algorithmic complexity by grounding the identification and quantification of evolved organization in measurable operations and multiplicities (Kempes et al., 2025).

To demonstrate this approach, we apply AT to the study of the *Universal Declaration of Human Rights* (UDHR) corpus (Vatanen et al., 2010), using characters as elementary units and concatenation as the joining operation defining assembly steps. Treating words as objects separated by the standard space delimiter, the English UDHR comprises $N = 536$ unique words drawn from an alphabet of 22 characters (including the space) and 1783 total words. Considering this natural tokenization, the text exhibits an Assembly value of $A \approx 1478$, reflecting nontrivial assembly indices for frequently reused substrings or words. We sweep every character as a possible delimiter and recompute A, both for the original text and 1000 randomized controls per delimiter, generated by shuffling characters to preserve text length and marginal letter frequencies. As seen in **Fig. 1**, the space delimiter emerges as a distinguished symbol: the resulting A is large (but not maximal) and robustly separated from randomized controls. Extending this analysis to over 100 UDHR languages, the most clearly separated distribution (which is also consistently the narrowest) tends to correspond to using the language's most frequent character as the delimiter—sometimes the space, but often a vowel such as *e* or *a*, particularly in languages with longer average word lengths. In addition, although the absolute Assembly value is strongly text dependent, the control distributions appear largely language-wide: for example, applying the same procedure to another English text (*Green Eggs and Ham*; 799 total words and 53 unique words) gives $A \approx 22$, yet the randomized-delimiter distributions remain centered at similar values and across all languages examined never exceed $A = 1$. Therefore, the Assembly of the original text is robustly distinguished from randomized strings regardless of delimiter choice, while the interpretable optimum in the original text reflects a balance between reuse and modularity that does not drive A to artificially extreme values. These results show that AT enables detecting evolved and selected linguistic structures, grounded in the recursive nature of language and the reuse of meaningful subparts, e.g., words, morphemes, recurrent phrases (Florence, 1950). In this context, language appears to occupy an intermediate and regulated complexity space between trivial repetition (low a, high n) and unstructured randomness (low n, high a), consistent with the optimization of learnability and meaning transmission under cognitive and cultural constraints (Fujita & Fujita, 2022; Levshina & Moran, 2021). The delimiter-agnostic nature of this approach has promising implications for undeciphered scripts, emergent machine codes and extraterrestrial communication, where the detection of an evolved alien signal heavily depends on data discretization (Tarter, 2001).

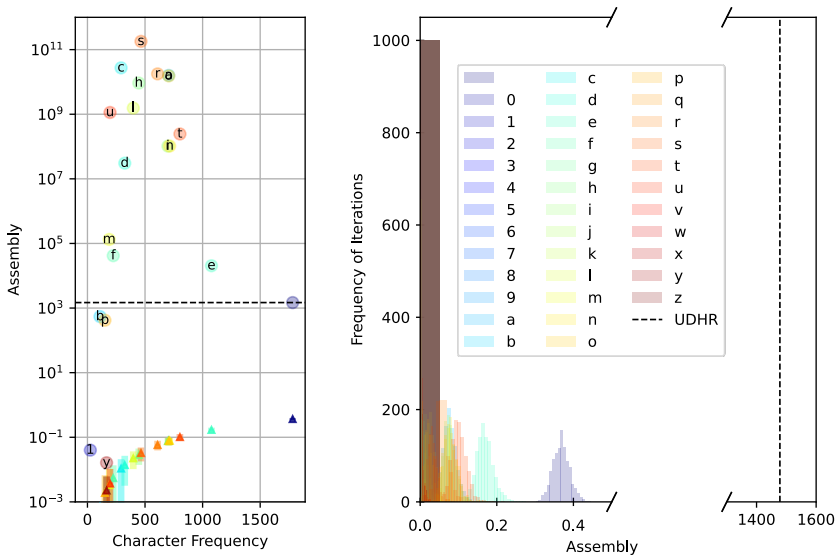

Figure 1: (Left) Assembly as a function of character frequency. The round data points correspond to A for the original English UDHR text, choosing different characters as the delimiter to define new words. The shaded vertical bands show the distributions of A for those same delimiters (their frequencies are unchanged) computed on randomized control versions of the text; the triangle markers mark the means of the distributions. (Right) Assembly distributions of 1000 randomized English UDHR strings for each delimiter. The original text is displayed as a black dotted line in both plots.

We present an extended, comparative analysis of string-based selection across multilingual literary corpora, quantifying how alphabet size, orthographic conventions, and tokenization rules modulate a-distributions and A, while tracing shifts across diachronic stages of each language. In parallel, we use assembly theory to interpret results from robotic experiments of language emergence and evolution (Vogt, 2015), in which embodied agents co-evolve an artificial language by combinatorically assembling phonemes and developing a shared vocabulary. Monitoring the understanding among agents via body language and signaling, coupled with computation of the complexity of their communicative outputs, enables identifying and characterizing structural phase transitions that mark the emergence of a common language. This dual research—corpus generalization and embodied experiments—provides a unified way to compare structure-driven diagnostics of selection and evolution across languages and modalities, while also building a bridge between artificial-life and origin-of-language studies.

References

Altmann, E. G., & Gerlach, M. (2016). Statistical Laws in Linguistics. In M. Degli Esposti, E. G. Altmann, & F. Pachet (Eds.), *Creativity and Universality in Language* (pp. 7–26). Springer International Publishing. https://doi.org/10.1007/978-3-319-24403-7_2

Florence, P. S. (1950). Human Behaviour and the Principle of Least Effort. *The Economic Journal*, *60*(240), 808–810. https://doi.org/10.2307/2226729

Fujita, H., & Fujita, K. (2022). Human language evolution: A view from theoretical linguistics on how syntax and the lexicon first came into being. *Primates*, *63*(5), 403–415. https://doi.org/10.1007/s10329-021-00891-0

Kempes, C. P., Lachmann, M., Iannaccone, A., Matthew Fricke, G., Redwan Chowdhury, M., Walker, S. I., & Cronin, L. (2025). Assembly theory and its relationship with computational complexity. *Npj Complexity*, *2*(1), 27. https://doi.org/10.1038/s44260-025-00049-9

Levshina, N., & Moran, S. (2021). Efficiency in human languages: Corpus evidence for universal principles. *Linguistics Vanguard*, *7*(s3). https://doi.org/10.1515/lingvan-2020-0081

Marshall, S. M., Mathis, C., Carrick, E., Keenan, G., Cooper, G. J. T., Graham, H., Craven, M., Gromski, P. S., Moore, D. G., Walker, S. I., & Cronin, L. (2021). Identifying molecules as biosignatures with assembly theory and mass spectrometry. *Nature Communications*, *12*(1), 3033. https://doi.org/10.1038/s41467-021-23258-x

Patarroyo, K. Y., Sharma, A., Seet, I., Packmore, I., Walker, S. I., & Cronin, L. (2025). *Quantifying the Complexity of Materials with Assembly Theory* (arXiv:2502.09750). arXiv. https://doi.org/10.48550/arXiv.2502.09750

Sharma, A., Czégel, D., Lachmann, M., Kempes, C. P., Walker, S. I., & Cronin, L. (2023). Assembly theory explains and quantifies selection and evolution. *Nature*, *622*(7982), 321–328. https://doi.org/10.1038/s41586-023-06600-9

Tarter, J. (2001). The Search for Extraterrestrial Intelligence (SETI). *Annual Review of Astronomy and Astrophysics*, *39*, 511–548. https://doi.org/10.1146/annurev.astro.39.1.511

Vatanen, T., Väyrynen, J. J., & Virpioja, S. (2010). Language Identification of Short Text Segments with N-gram Models. In N. Calzolari, K. Choukri, B. Maegaard, J. Mariani, J. Odijk, S. Piperidis, M. Rosner, & D. Tapias (Eds.), *Proceedings of the Seventh International Conference on Language Resources and Evaluation (LREC'10)*. European Language Resources Association (ELRA). https://aclanthology.org/L10-1193/

Vogt, P. (2015). How mobile robots can self-organise a vocabulary. In *Language Science Press*. Language Science Press. https://doi.org/10.17169/langsci.b50.113

Ziv, J., & Lempel, A. (1978). Compression of individual sequences via variable-rate coding. *IEEE Transactions on Information Theory*, *24*(5), 530–536. https://doi.org/10.1109/TIT.1978.1055934